5-Implementation of Quick Sort

Write a Program to Implement the Quick Sort Algorithm

Input Format:
The first line contains the no of elements in the list-n
The next n lines contain the elements.

Output:
Sorted list of elements

For example:

Input	Result
5 67 34 12 98 78	12 34 67 78 98

Answer:

Feedback

#include <stdio.h>

```
void quickSort(int arr[], int low, int high)
{
   if (low < high)
   {
      int i = low;
      int j = high;
      int pivot = arr[low];
      int temp;

      while (i < j)
      {
         while (arr[i] <= pivot && i < high)
            i++;

         while (arr[j] > pivot)
            j--;

         if (i < j)
         {
            temp = arr[i];
            arr[i] = arr[j];
            arr[j] = temp;
         }
      }

      temp = arr[low];
      arr[low] = arr[j];
```

```
        arr[j] = temp;

        quickSort(arr, low, j - 1);
        quickSort(arr, j + 1, high);
    }
}

int main()
{
    int n;

    scanf("%d", &n);

    int arr[n];

    for (int i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    quickSort(arr, 0, n - 1);

    for (int i = 0; i < n; i++)
    {
        printf("%d", arr[i]);

        if (i < n - 1)
            printf(" ");
    }
```

```
    return 0;
}
```

	Input	Expected	Got	
	5 67 34 12 98 78	12 34 67 78 98	12 34 67 78 98	
	10 1 56 78 90 32 56 11 10 90 114	1 10 11 32 56 56 78 90 90 114	1 10 11 32 56 56 78 90 90 114	
	12 9 8 7 6 5 4 3 2 1 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	

Passed all tests!

Correct

Marks for this submission: 1.00/1.00.

Dynamic Programming: